

Artificial Intelligence & Machine Learning

Questions and Answers

Q1. How does AI differ from traditional programming?

Ans: In traditional programming, developers write explicit rules and the computer follows them to produce output. In AI, the system learns patterns from data and builds its own rules. Traditional programs fail on situations not coded for, while AI can generalize to new inputs and improve with more data.

Q2. What are some real-world applications of AI?

Ans: AI is used in healthcare for disease detection and drug discovery, in finance for fraud detection, and in transport for self-driving cars. It also powers virtual assistants like Siri and Alexa, recommendation systems on Netflix and Amazon, and language translation tools.

Q3. Explain the difference between Narrow AI, General AI and Super AI.

Ans: Narrow AI is designed for one specific task, such as face recognition or chess, and cannot go beyond it. General AI would match human intelligence and be able to learn and reason across any domain. Super AI is a hypothetical level that would exceed human intelligence in every field. Only Narrow AI exists today.

Q4. Explain how Super AI could potentially surpass human intelligence.

Ans: A Super AI could process and store vast amounts of data at speeds far beyond the human brain. It could improve its own algorithms repeatedly, a process called recursive self-improvement, leading to rapid growth in ability. It would also work nonstop without fatigue, memory limits, or emotional bias.

Q5. What ethical concerns are associated with Super AI?

Ans: Major concerns include loss of human control, since a Super AI's goals may not align with human values. Other worries are mass unemployment, misuse for surveillance or weapons, and unfair decisions caused by bias. There is also the question of who is accountable when a highly autonomous system causes harm.

Q6. What is Reactive Machine AI? Give examples.

Ans: Reactive Machine AI is the simplest type of AI. It responds only to the current input and has no memory of past events or ability to learn from them. Examples include IBM's Deep Blue chess computer and basic spam filters that act only on the present data.

Q7. Explain the concept of Theory of Mind AI.

Ans: Theory of Mind AI is a future type of AI that could understand human emotions, beliefs, intentions, and thoughts. It would adjust its behavior based on how people feel and what they expect. This would allow natural social interaction, but the technology is still under research and does not exist yet.

Q8. How does Limited Memory AI differ from Reactive AI?

Ans: Limited Memory AI can store past data for a short time and use it to make better decisions, while Reactive AI cannot remember anything. For example, a self-driving car uses recent observations of speed and nearby vehicles to decide its next move. Most modern AI systems fall in the Limited Memory category.

Q9. How does Machine Learning relate to Artificial Intelligence?

Ans: Machine Learning is a subset of Artificial Intelligence. AI is the broad goal of making machines act intelligently, and ML is one approach to achieve it by letting machines learn from data. Deep Learning is in turn a further subset of ML that uses neural networks.

Q10. Give examples of Machine Learning applications in daily life.

Ans: ML powers movie and product recommendations on Netflix and Amazon, and spam filtering in email. It enables voice assistants, face unlock on phones, and predictive text on keyboards. Navigation apps like Google Maps also use it to estimate traffic and travel time.

Q11. What is the difference between training and testing data?

Ans: Training data is the dataset used to teach the model so it can learn patterns. Testing data is a separate set the model has never seen, used to check how well it performs on new inputs. Keeping them separate gives an honest measure of accuracy and helps detect overfitting.

Q12. Which learning method is best suited for labeled data?

Ans: Supervised learning is best suited for labeled data. The model learns from input-output pairs, where each example has a known correct answer. It then uses this knowledge to predict labels for new data, as in email spam detection or house price prediction.

Q13. Compare Supervised and Unsupervised Learning.

Ans: Supervised learning uses labeled data and aims to predict known outputs, using methods like classification and regression. Unsupervised learning uses unlabeled data and tries to find hidden patterns or groups on its own, such as clustering. Supervised results are easier to evaluate, while unsupervised learning is useful for exploring data.

Q14. Explain the difference between classification and regression.

Ans: Classification predicts a discrete category or class, such as whether an email is spam or not spam. Regression predicts a continuous numerical value, such as the price of a house or tomorrow's temperature. Both are supervised learning tasks, but they differ in the type of output.

Q15. What is overfitting in supervised learning?

Ans: Overfitting happens when a model learns the training data too well, including its noise and random details. As a result it scores high on training data but performs poorly on new, unseen data. It can be reduced using more data, simpler models, regularization, or cross-validation.

Q16. How does unsupervised learning differ from supervised learning?

Ans: Unsupervised learning works with data that has no labels and no correct answers provided. Instead of predicting outputs, it discovers structure such as clusters, associations, or patterns. Supervised learning, in contrast, is guided by labeled examples and a clear target to predict.

Q17. What is dimensionality reduction?

Ans: Dimensionality reduction is the process of reducing the number of features or variables in a dataset while keeping the most important information. It makes models faster, reduces noise, and helps visualize high-dimensional data. Common techniques include PCA (Principal Component Analysis) and t-SNE.

Q18. How does reinforcement learning differ from supervised learning?

Ans: In supervised learning, the model learns from a dataset with correct answers given in advance. In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties for its actions. There are no fixed labels, so it learns through trial and error to maximize long-term reward.

Q19. Explain the concept of "reward" in RL.

Ans: A reward is a numerical feedback signal the environment gives an agent after it takes an action. Positive rewards encourage good actions and negative rewards (penalties) discourage bad ones. The agent's goal is to learn a strategy that maximizes the total reward over time, such as scoring points in a game.

Q20. What are some challenges in unsupervised learning?

Ans: Since there are no labels, it is hard to measure whether the results are correct or useful. Choosing the right number of clusters and the right algorithm is often difficult. The results can be hard to interpret, and the methods are sensitive to noise, outliers, and high-dimensional data.